

The empirical recurrence for OEIS A277769: recovering a conjecture that is not written down

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Abstract

OEIS A277769 records “Empirical recurrence of order 98”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 134$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 98, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 98 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A277769 is “Number of $4 \times n$ 0..2 arrays with every element equal to some element at offset (-1,-1) (-1,0) (-1,1) (0,-1) (0,1) or (1,0) both plus 1 mod 3 and minus 1 mod 3, with new values introduced in order 0..2.”. It has offset 1 and begins

$$0, 14, 330, 9741, 275018, 7824415, 221983169, 6298326387, \dots$$

The entry states:

Empirical recurrence of order 98 (see link above)

As of the “Last modified” line on the live entry (Oct 29 2016 07:32:40, revision 4) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 134$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 134$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 268$ exact terms of the model returns order 98 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{98} c_i a(n-i),$$

c_1	29	c_{26}	−1035437875193547539	c_{51}	897502215710441873981	c_{76}	6942410380
c_2	39	c_{27}	2093027498779176207	c_{52}	844075578025670186319	c_{77}	−251685945
c_3	−338	c_{28}	−5056425154521149491	c_{53}	−1958751446712713439454	c_{78}	−759050935
c_4	−38872	c_{29}	10050838076250006577	c_{54}	3037330425072980423793	c_{79}	2697066566
c_5	14183	c_{30}	−19194891713877055722	c_{55}	−3563519167393715392762	c_{80}	−36063956
c_6	1292572	c_{31}	36598861831513059630	c_{56}	3847824360567607093747	c_{81}	629604972
c_7	17923186	c_{32}	−68134069823052811722	c_{57}	−4748301350199262878344	c_{82}	−44030818
c_8	−43150484	c_{33}	116584983780648698618	c_{58}	4001535737541730603969	c_{83}	155638759
c_9	−540346910	c_{34}	−191239964128297455999	c_{59}	−3215719980856614347251	c_{84}	−23129334
c_{10}	−3501463942	c_{35}	321386238836827998998	c_{60}	3713268352773232483270	c_{85}	14371134
c_{11}	15166603708	c_{36}	−499656386714429100525	c_{61}	−2691958591260098474529	c_{86}	−12669025
c_{12}	76455111243	c_{37}	720080721918713439256	c_{62}	1532116980054811688188	c_{87}	−2644605
c_{13}	334637207006	c_{38}	−1065922519428335996475	c_{63}	−1767096470318537125731	c_{88}	18848530
c_{14}	−1859521801781	c_{39}	1538061984853459373663	c_{64}	1235022024289208338249	c_{89}	−1465486
c_{15}	−3090361845866	c_{40}	−1919380134402065672144	c_{65}	−420347788967071462648	c_{90}	2501356
c_{16}	−24852237063288	c_{41}	2497381177310941467003	c_{66}	521840844907278593357	c_{91}	−809889
c_{17}	100817482803800	c_{42}	−3337816590659106451124	c_{67}	−418360723176360695455	c_{92}	−175157
c_{18}	−7014175734417	c_{43}	3528079372270821719265	c_{68}	64168714975072092944	c_{93}	614582
c_{19}	1166166985188053	c_{44}	−4011967102148614152896	c_{69}	−84822635503573088019	c_{94}	182202
c_{20}	−3453635016649445	c_{45}	4876982021268174743894	c_{70}	105462967951299630936	c_{95}	−13423
c_{21}	4351146909422436	c_{46}	−4195809233522677513261	c_{71}	−6578280825999003829	c_{96}	60983
c_{22}	−30458720798281225	c_{47}	4002222766248328387177	c_{72}	−592125380572110356	c_{97}	−2842
c_{23}	73635533140762924	c_{48}	−4251514821611735049920	c_{73}	−16493465173638871578	c_{98}	−5951
c_{24}	−137070463474432206	c_{49}	2529355636185779869266	c_{74}	1116965187132769931		
c_{25}	485288581225450537	c_{50}	−1528287652132014723092	c_{75}	3779496022249733441		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 98, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $98 + 4$ terms removed returns order 98 as well, so $\deg q_{\min} = 98 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 16 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $98 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 98 that this sequence satisfies — and that family turns out to have exactly one member.

References

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